Kerb Mass Reduction Roadmap

HCV rigid platform, FY24 to FY27

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Approved by	Programme Director - HCV Rigid
Baseline	FY24 build, 2,840 kg kerb mass, 28 t GVW rigid
Target	2,608 kg by FY27, a reduction of 232 kg
Distribution	Programme Board; VAVE Committee; Product Engineering

Summary

The platform carries a committed kerb-mass reduction of 232 kg between FY24 and FY27, equivalent to 8.2 per cent. This roadmap allocates that target across five sub-system streams, records the ideas currently committed against each, and identifies the gap still to be filled. As at the mid-year review, 171 kg is committed against approved ideas and 61 kg remains unallocated, concentrated in the chassis frame stream.

Contents

1	Target and basis	2
2	Allocation by sub-system	2
3	Committed ideas	4
4	The unallocated gap	5
5	Dependencies and risks	5

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1 Target and basis

The FY24 baseline kerb mass of 2,840 kg is the measured average of twelve production vehicles built in September 2024, in the standard specification with a full fuel tank and without body. The FY27 target of 2,608 kg is derived from the payload commitment in the platform brief and is not negotiable at sub-system level.

1.1 Why mass, and why now

Kerb mass converts directly to payload at a fixed GVW, and payload is the primary purchase criterion in this segment. A 232 kg reduction is worth approximately 0.8 per cent of payload on a 28 tonne rigid. It also reduces unladen fuel consumption by an estimated 1.4 per cent, which affects the total cost of ownership case.

> Basis note. Mass reductions in this roadmap are counted at the vehicle, not at the component. A 2 kg saving on a part fitted twice counts as 4 kg. Savings on parts not fitted to the standard specification do not count at all.

2 Allocation by sub-system

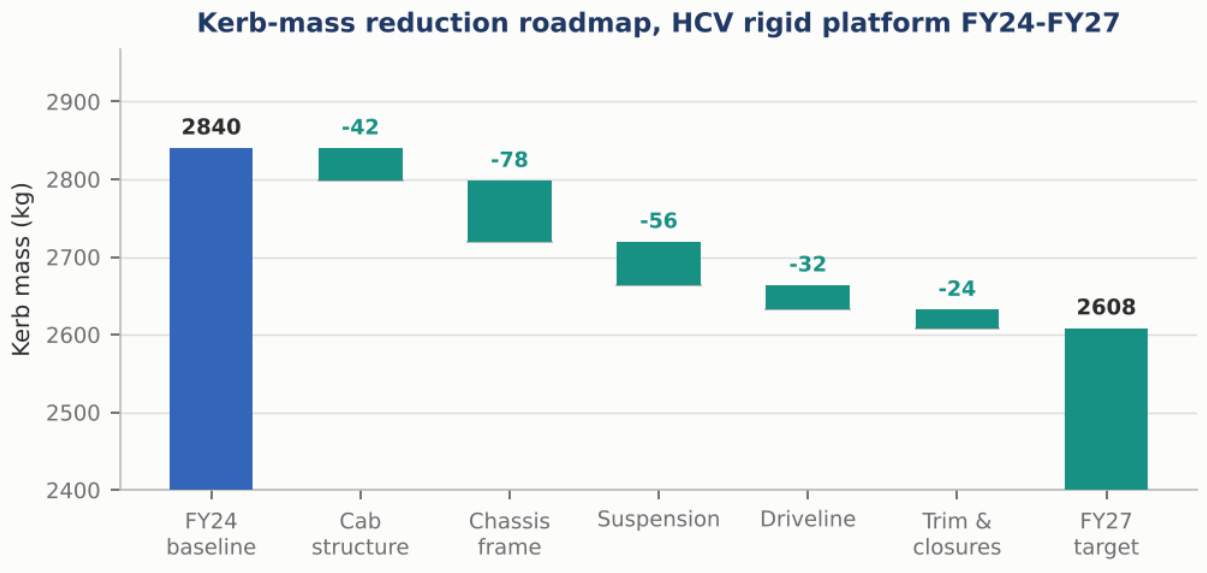


Figure 1 - Kerb mass reduction roadmap by sub-system stream. The chassis frame carries the largest single allocation at 78 kg.

Stream	Allocation (kg)	Committed (kg)	Gap (kg)	Confidence
Cab structure	42	38	4	High
Chassis frame	78	41	37	Low
Suspension	56	48	8	Medium
Driveline	32	26	6	Medium
Trim and closures	24	18	6	High
Total	232	171	61	

Finding 2.1. Sixty-one per cent of the remaining gap sits in the chassis frame stream, which also carries the lowest confidence. The frame is the stream with the longest validation lead time, so the gap there is the one least able to absorb a late recovery.

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3 Committed ideas

Idea ref	Change	Stream	Mass (kg)	Status
VC-24-0129	5 mm crossmember with hanger doubler	Chassis frame	18.4	Implemented
VC-24-0301	Tapered mid-span front axle beam	Suspension	19.0	In validation
VC-24-0188	Pressed brake chamber bracket	Chassis frame	5.68	Approved
VC-24-0066	PMMA lens, low-impact positions	Trim and closures	1.84	Implemented
VC-23-0219	Door hinge bracket web deletion	Cab structure	2.56	Implemented
VC-24-0233	Aluminium exhaust heat shield	Driveline	4.20	Approved
VC-24-0411	Parabolic rear spring, 3 to 2 leaf	Suspension	22.6	In validation
VC-24-0455	Composite air reservoir	Chassis frame	8.90	In validation
VC-24-0502	Headliner substrate to PET felt	Cab structure	3.40	On hold
VC-24-0188 b	Cab rear panel gauge reduction	Cab structure	11.2	Approved

3.1 Ideas at risk

Idea ref	Risk	Mass at risk (kg)
VC-24-0502	Hot-climate sag complaints on the PET felt substrate; reversal under consideration	3.40
VC-24-0411	Ride acceptance not yet demonstrated at part load	22.6
VC-24-0455	Supplier capability not confirmed for the composite reservoir	8.90

Finding 3.1. Thirty-four point nine kilograms of the 171 kg committed is carried by three ideas with unresolved risk. If all three fail, the gap rises from 61 kg to 96 kg, which would not be recoverable within the FY27 timing.

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4 The unallocated gap

Sixty-one kilograms remain unallocated. The candidate pool below has been screened for technical feasibility but not costed or validated. It is recorded so that the gap is visible rather than assumed to be closable.

Candidate	Stream	Indicative mass (kg)	Principal obstacle
High-strength frame side member	Chassis frame	26.0	Requires frame re-homologation
Aluminium fuel tank straps and brackets	Chassis frame	6.4	Dissimilar-material joint, see ES-STD section 4.2
Composite propeller shaft	Driveline	9.8	Critical speed at the longest wheelbase
Cab floor gauge reduction	Cab structure	7.2	Crash performance not assessed
Single-piece moulded dash carrier	Trim and closures	5.6	Tooling lead time exceeds FY27 timing
Hollow stabiliser bar, front	Suspension	4.8	Supplier capability not on panel

The candidate pool totals 59.8 kg against a gap of 61 kg, with no margin. Every candidate carries a material obstacle and two of them, the frame side member and the dash carrier, cannot be delivered within FY27 timing even if approved immediately.

Finding 4.1. The roadmap is not closable from the current candidate pool. Either the target is re-phased, or ideas must be generated against the chassis frame stream specifically. The idea register contains 41 closed frame proposals that have not been re-tested against the current mass target.

5 Dependencies and risks

- Validation capacity. Four of the committed ideas require frame-level durability testing and the rig is at capacity until Q3 FY26.
- Reversal risk. Two committed ideas are in the same component family as a change reversed in FY24; see CVOEM/VAVE/WF-FA/2024/021.
- Cost interaction. Mass reduction and cost reduction are both committed and several candidates deliver one at the expense of the other. The frame side member saves 26 kg and adds INR 1,840.
- Supplier lead time. Three candidates require new hard tooling with lead times of 26 to 34 weeks; see CVOEM/VAVE/SC-TC/2024/009 section 1.1.

Recommendation. Re-test the 41 closed chassis frame proposals against the current mass target before concluding that the gap requires new ideas. Several were closed on cost grounds at material rates that have since moved.

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